

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with 8+ years in banking and capital markets, building Python and Java services, real-time data platforms, and event-driven systems with Apache Kafka. Applies AI agents to accelerate migration, reconciliation, and workflow automation while modernizing legacy OLTP and batch pipelines toward cloud-native, highly available architectures. Strong CI/CD, cross-team collaboration, and regulated-industry delivery from design through production operations.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected **real-time data ingestion services** replacing legacy batch file transfers with **event-driven Apache Kafka** pipelines feeding an internal OLTP datastore, improving downstream freshness for portfolio analytics teams.
- Built **Python and Java REST APIs** supporting capital operations workflows, maintaining **99.9% uptime** across distributed services deployed through automated CI/CD pipelines.
- Led **AI agent-assisted migration tooling** to reconcile source-to-target datasets during platform modernization, reducing manual validation effort for data stewards by an estimated **40%**.
- Delivered **full stack capabilities** with React and TypeScript frontends backed by scalable microservices, enabling operators to monitor migration status and exception queues in real time.
- Partnered with architecture, security, and networking teams to implement **cloud-native deployment patterns**, configuration management, and secrets handling for regulated financial workloads.
- Implemented **observability dashboards and alerting** for high-availability services, shortening mean time to detect production incidents across portfolio reporting pipelines.
- Mentored engineers on **event-driven design**, API contracts, and testing strategies while contributing to cross-team standards for continuous delivery and code review.
- Automated **operational reconciliation workflows**, integrating batch validation steps into streaming consumers to improve efficiency for back-office teams.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java and Python backend services** supporting trading and banking operations on high-volume OLTP systems, processing **millions of daily transactions** with strict latency requirements.
- Migrated **legacy batch file interfaces** to modern streaming ingestion patterns, coordinating phased cutover plans for operational data stores with data services teams.
- Built **internal web applications** for technology analysts to track change management, deployment windows, and production incidents across distributed banking platforms.
- Enhanced **CI/CD pipelines** and configuration management with platform teams, adding automated test gates for continuous integration in a regulated banking environment.
- Supported **data migration initiatives** between core banking systems and downstream analytics repositories, validating record counts and schema mappings through scripted reconciliation.
- Implemented **Kafka-based event producers and consumers** for real-time status updates across operational workflows, improving visibility for operations and control teams.
- Contributed to **networking and security reviews** for service-to-service communication, ensuring compliance with enterprise banking standards for encryption and access control.
- Created **Python automation tooling** for log analysis, batch monitoring, and alert routing, improving operational efficiency during production support rotations.

- Participated in **cross-functional modernization workstreams** replacing batch integrations with API-first patterns while maintaining high availability during migration windows.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React components and REST integrations** for an internal content management portal, improving editor workflow efficiency for product stakeholders.
- Developed **Python scripts** to automate data validation and reporting tasks, reducing manual checks during feature release cycles.
- Collaborated with senior engineers on **CI/CD setup** for frontend and backend repositories, enabling automated builds and deployment to staging environments.
- Implemented **Node.js API endpoints** supporting user authentication and document retrieval with unit test coverage for core business logic.
- Assisted with **database query optimization** and indexing changes that improved page load performance for customer-facing features.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** with HTML, CSS, and JavaScript for foundation outreach programs, ensuring accessibility across mobile and desktop browsers.
- Integrated **third-party form and CMS tools** to streamline content updates for non-technical staff without developer intervention.
- Maintained **cross-browser compatibility** and performance tuning for public-facing sites serving community program information.
- Collaborated with stakeholders to translate **workflow requirements** into lightweight web features delivered within tight nonprofit timelines.
- Deployed site updates through a **version-controlled release process**, documenting changes for volunteer maintainers.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Responsive Web Design, Component Architecture

Backend: Java, Python, Node.js, REST APIs, Microservices, Event-Driven Architecture, Apache Kafka

Databases: PostgreSQL, MySQL, Oracle, SQL Server, OLTP Data Modeling, Schema Migration, Snowflake

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Terraform, Configuration Management

Monitoring & Tools: Prometheus, Grafana, Datadog, Splunk, ELK Stack, JUnit, pytest, AI Agent Frameworks

Workflow & PM: Agile, Scrum, Jira, Confluence, Change Management, Cross-Functional Collaboration